CR23532 – CYBER AND DIGITAL FORENSICS

CASE STUDY ASSIGNMENTS – II

B.E. CSE – Cyber Security

Rajalakshmi Engineering College

Assignment 2(1)

Case Study: Suspected Unauthorized Access to a College Server

Introduction and Assumptions

This case concerns the suspected access and copying of confidential examination-related files from a college departmental server outside normal working hours. The investigation must determine what happened, when it happened, which account or device was involved, and whether the available evidence supports the allegation. The discussion below describes a proposed forensic workflow; it does not claim that any tool was actually run or that the suspect has been proven responsible. It is assumed that the college can provide written authorization and that relevant server logs and devices remain available.

1. Explain the Identification, Collection, Examination, Analysis, and Presentation phases.

Identification is the process of locating possible evidence and defining the scope of the incident. Investigators should record the administrator's complaint, the affected server and folders, the suspected time period, users with access, relevant workstations, removable media, and systems that may hold authentication or network logs. They should identify who owns each system and obtain authorization before accessing it. Volatile information, such as active sessions, running processes, and network connections, may need to be considered promptly because it can disappear when a system is shut down.

Collection involves securing the scene and acquiring the identified evidence in a controlled manner. Investigators should photograph and document devices, record serial numbers and system details, export relevant logs using approved procedures, and create forensic images of storage media where appropriate. A write blocker should be used when imaging compatible storage. Original evidence should be preserved, and every transfer should be documented in a chain-of-custody record.

Examination is the technical extraction and organization of information from collected evidence. Investigators can review server access logs, authentication records, file-system metadata, recent-file records, USB connection artifacts, and relevant browser or command history. Searches should be focused on the suspected period and the examination should preserve original timestamps and metadata.

Analysis interprets the extracted artifacts together. The team should correlate account logins, file access events, possible copying activity, USB connection times, and workstation activity. It should test alternative explanations such as scheduled backups, authorized staff activity, clock differences, or shared accounts. A single log entry should not be treated as proof of a person's actions without corroboration.

Presentation communicates the methods, evidence, results, limitations, and conclusions in a clear report. The report should distinguish facts directly supported by artifacts from interpretations and unresolved questions. Relevant exhibits, hash values, and chain-of-custody records should be retained so another qualified examiner can review the work.

2. Prepare a forensic investigation flowchart showing the sequence of activities.

Because this submission is presented in a normal written format without a flowchart, the sequence is described in words. First, receive and document the complaint. Next, obtain authorization and define the scope. Identify potential evidence and assess whether volatile data must be collected. Secure the devices and logs, document their condition, and acquire forensic copies where appropriate. Verify acquisitions with cryptographic hashes and record custody transfers. Examine the copies for relevant logs, metadata, account activity, and removable-media artifacts. Correlate the findings into a timeline, test alternative explanations, and validate important results. Finally, prepare the report, preserve exhibits, and present the conclusions to the authorized college authority.

3. Identify at least five items of potential digital evidence and classify their investigative value.

The administrator's complaint is contextual evidence because it defines the allegation, affected files, and approximate time period, but it does not independently prove access. Server access and authentication logs are potentially direct technical evidence of a particular account, session, or file request, depending on the detail and reliability of the logs. The suspect workstation's forensic image may contain user activity, recent-file records, shell history, event logs, and traces of file transfers; its value depends on whether the artifacts can be tied to the relevant period and account. The recovered USB device may contain copied files, file metadata, volume identifiers, and remnants of deleted data, providing evidence of possible transfer. Relevant examination-file metadata can show creation, modification, and access information, although timestamps may be altered or affected by copying. Network or firewall logs may corroborate connections between a workstation and the server. Account-permission records can show whether an account had the ability to access the files, but permission alone does not show that access occurred. The strongest conclusion comes from multiple independent artifacts that agree.

4. Explain precautions during collection and preservation.

Investigators should have proper legal or institutional authorization and collect only evidence within the approved scope. They should photograph the scene, note date, time, location, device condition, connected peripherals, and visible screen state before changing anything. They should avoid browsing files manually on the original system, because doing so can change timestamps or other data. If volatile data must be collected, the examiner should document each action and its reason. Storage should be acquired with a validated tool and write blocker when possible. Original devices

should be sealed, labelled, and stored securely; analysis should be performed on verified forensic copies. The examiner should calculate cryptographic hashes, such as SHA-256, before and after acquisition where supported, and record tool versions, settings, errors, and operator details. Every transfer, opening of a seal, and return to storage should be documented. Evidence should be protected from heat, moisture, unauthorized access, and accidental alteration.

5. Describe how investigators can establish a timeline.

The team should collect relevant timestamps from server logs, authentication records, workstation event logs, file-system metadata, USB artifacts, and network records. All times should be normalized to a common reference, preferably UTC, while preserving the original timestamp and the system's configured time zone. Investigators should check whether device clocks were synchronized and document any known offset. Events can then be ordered by time: an account login, a request to access an examination folder, a file read or download, a USB device connection, and evidence of the file appearing on the USB. The timeline should identify the source of each event and its reliability. If timestamps conflict, the discrepancy should be investigated rather than silently corrected. A sequence of consistent artifacts can support a reconstruction, but a timeline alone does not automatically identify the human operating the account.

6. Explain what should be included in the final report and how findings should be presented to a non-technical authority.

The report should contain the case identifier, requesting authority, examiner details, scope and authorization, incident summary, evidence inventory, chain-of-custody details, acquisition methods, tool names and versions, hash values, examination steps, findings, timeline, interpretation, limitations, and conclusion. Important findings should be supported by references to logs, screenshots, extracted records, or other exhibits. The report should state whether the evidence supports, does not support, or is insufficient to resolve each allegation. For a non-technical audience, explain terms in plain language. For example, state that a server record shows a particular account requested a file at a particular time, rather than assuming the reader understands log fields. Separate confirmed facts from possible explanations and avoid claiming that a named individual performed an action unless the evidence supports that attribution.

7. Discuss legal and ethical considerations in a college environment.

Investigators should follow applicable law, institutional policy, privacy requirements, and the written scope of authorization. They should avoid searching unrelated student or staff information merely because it is technically accessible. Personal data encountered during the investigation should be handled confidentially and disclosed only to authorized recipients. The process must be impartial: investigators should preserve both incriminating and exculpatory evidence, document uncertainty, and avoid prejudging the suspect. Access to evidence should be restricted, and copies should be stored securely. If the investigation may lead to disciplinary or legal action, the college should consult its authorized legal or compliance personnel to ensure that collection and reporting requirements are met.

Assignment 2(2)

Case Study: Linking a Suspicious File to a User Using File-System Evidence

Introduction and Assumptions

Several staff members use a shared desktop containing a suspicious executable, hidden files, encrypted documents, browser artifacts, and multiple user accounts. The objective is to determine whether the evidence supports a link between a particular account and the suspicious activity. The actual identity of the person responsible cannot be assumed from the presence of a file alone. ILookIX is discussed as the proposed forensic analysis tool, and no claim is made that an examination has already been performed.

8. Identify the hardware and software components that should be documented before analysis.

The examiner should record the computer manufacturer, model, asset tag, serial number, CPU, RAM, internal storage type and capacity, connected external drives, USB devices, network interfaces, and any relevant peripherals. For each storage device, record its make, model, serial number, capacity, interface, and physical condition. The software inventory should include operating system edition and version, file-system types, installed applications, user accounts, security tools, encryption status, system time zone, clock settings, and relevant browser versions. Investigators should note whether the machine was powered on or off and whether it was connected to a network. Photographs and a written inventory help establish what was examined and allow the setup to be reproduced.

9. Explain how storage devices, operating systems, file systems, and metadata help locate evidence.

Storage devices retain allocated files and may also contain deleted data, unallocated space, file-system structures, and remnants of previous activity. The operating system records user logons, application execution, device connections, and system events. File systems such as NTFS organize files through structures that can preserve file names, paths, timestamps, permissions, and references to file content. Metadata can reveal when a file was created or modified, its size, ownership or security permissions, and sometimes the application associated with it. These details can help locate suspicious items and compare them with account activity. However, timestamps may change during copying, can be manipulated, and may reflect system processes rather than deliberate user actions. Findings should therefore be correlated with other artifacts.

10. Describe how hidden and encrypted files affect investigation and appropriate handling.

Hidden files may be excluded from ordinary directory views but can still be located through forensic tools that enumerate file-system entries, search paths, and examine metadata. Investigators should record the original path, attributes, timestamps, and any links to user activity. Encryption may prevent the content from being read without a key or password. Examiners should preserve the encrypted original, identify the encryption type where possible, and use only legally authorized methods to obtain access. They should not alter the original file or guess passwords in a way that

risks modifying evidence. A file being hidden or encrypted is not proof of wrongdoing; legitimate software and users also use these features.

11. Explain how file names, paths, timestamps, accounts, metadata, and other artifacts can be correlated to a user.

Investigators should begin with the suspicious file's full path, name, size, cryptographic hash, file-system metadata, and security permissions. They should determine which account could access the containing directory and whether account-specific artifacts show interaction with the file. Relevant evidence may include recent-file lists, shortcut files, prefetch or application-execution artifacts where available, shell or command history, event logs, browser downloads, and USB connection records. The examiner can compare the file's timestamps with logon sessions and other activity. A consistent match between an account session, an execution artifact, and a file operation provides stronger support than a file name or timestamp by itself. The correlation should be documented with the source and limitations of each artifact.

12. Explain a suitable workflow for importing/acquiring evidence and analyzing it using ILookIX.

First, obtain authorization and document the system and evidence identifiers. If a physical device is available, acquire a forensic image using an appropriate validated acquisition method and write protection where applicable. Calculate and record hash values and preserve the original media. In the ILookIX workflow, create or open the authorized case, enter case and evidence details, and import the forensic image or supported evidence source without changing the original. Verify the integrity information available from the acquisition. Next, examine file-system structures, enumerate user profiles, search for the suspicious executable and hidden files, inspect metadata, and review relevant operating-system and browser artifacts. Use keyword and hash searches where appropriate, and export only relevant findings with source references. Record tool version, settings, errors, and timestamps. The results should be checked against independent artifacts and documented in a report. Exact menu names and supported formats may vary by ILookIX version, so the examiner should follow the installed version's documentation.

13. Differentiate a direct link from supporting or contextual information.

A direct technical link is an artifact that records a specific activity in a way closely connected to the event, such as an authenticated session log showing a file access or a reliable application record showing execution under a particular account. Even then, it usually links the event to an account or device, not conclusively to the person at the keyboard. Supporting or contextual evidence includes file permissions, a user's general access to a folder, a file being present in a profile, or a system clock setting. Such evidence helps explain the circumstances but does not establish that the user performed the activity. Strong attribution requires independent corroboration and consideration of shared access, remote control, malware, and other alternatives.

14. Identify limitations or alternative explanations before attributing activity to a user.

First, the desktop is shared, so another staff member may have used the same device or left a session open. Second, account credentials may have been shared, stolen, or used remotely. Third, malware or an automated process may have created or executed the file without the account holder's

knowledge. Fourth, file timestamps may be affected by copying, clock drift, time-zone settings, or deliberate modification. Fifth, the presence of a file in a user profile may result from a download, synchronization, or another user's action. Sixth, encryption or deleted artifacts may leave gaps in the evidence. Investigators should state these limitations, seek corroboration, and avoid conclusions stronger than the available evidence.

Assignment 2(3)

Case Study: Forensic Imaging and Safekeeping of a Suspect Storage Device

Introduction and Assumptions

A laptop and an attached USB drive have been seized in an investigation into suspected data theft. The priority is to preserve original media and create verifiable forensic copies. IXImager and the ASB container format are considered as part of the proposed workflow. Procedures must be adapted to the actual tool version and evidence type; this assignment describes an intended process rather than claiming that imaging was carried out.

15. Prepare a step-by-step procedure for collecting and transporting the laptop and USB device.

Obtain the appropriate authorization and record who collected the devices, where and when they were found, and the reason for seizure. Photograph the laptop, screen, ports, connected USB device, cables, and surrounding setup before moving anything. Record whether the laptop is powered on, asleep, or off. If it is powered on, do not casually shut it down; consult the forensic lead because volatile data or encryption keys may be lost. Document any decision to collect live data. Label the laptop and USB separately and record make, model, serial number, and visible condition. If the USB is connected, document its port and connection before removing it under a controlled procedure. Use suitable antistatic packaging and tamper-evident seals, keep devices away from heat, moisture, magnetic hazards where relevant, and unauthorized access, and transport them securely. Record every handover in the chain of custody.

16. Explain physical acquisition and why analysis should normally use a forensic image.

Physical acquisition aims to capture the accessible contents of a storage device at the physical or sector level, including allocated files, file-system structures, unallocated space, and areas that may contain remnants of deleted files, subject to device and tool limitations. A forensic image creates a stable working copy of the source. Analysis on the image reduces the risk of modifying original evidence and allows repeatable examination by other investigators. It also supports verification through cryptographic hashes. Some modern devices, such as SSDs with TRIM or encrypted storage, may not expose all previously deleted data, so physical acquisition does not guarantee complete recovery. The original should be preserved securely while examination proceeds on verified copies.

17. Design a forensic imaging workflow using IXImager.

Begin by documenting the case, operator, source device identifiers, capacity, connection method, and physical condition. Confirm that the selected source is the suspect device and that the destination has sufficient capacity. Connect the source through a suitable hardware write blocker where supported, and verify that the source is not being mounted for ordinary write access. In IXImager, select the correct source and destination, choose a supported forensic image format and appropriate acquisition options, and enter case identifiers and examiner details. Start acquisition and monitor the process for read errors or interruptions. Preserve the tool's logs and note any unreadable

sectors rather than hiding errors. After acquisition, calculate or verify cryptographic hashes using the tool or an independent utility. Compare source and image verification values when the acquisition method supports a meaningful comparison. Store the image securely, mark it as an evidence master, and conduct analysis on a verified working copy. The exact options should be confirmed in the installed IXImager documentation.

18. Explain how hash values demonstrate integrity before and after acquisition.

A cryptographic hash, such as SHA-256, produces a fixed-length digest from the contents being hashed. If the same data is hashed using the same algorithm, the result should match; an unintended content change will normally produce a different digest. Investigators should record the algorithm, hash value, tool, date, and evidence identifier. Where supported, they should compare the acquisition verification value with a later verification value for the image. Matching values provide strong evidence that the hashed content has remained unchanged since the earlier calculation. A hash does not prove that the source was authentic before collection or that the acquisition captured every possible physical area; it demonstrates integrity of the data that was actually hashed.

19. Describe the role of an ASB container and the information that should accompany evidence.

An ASB container, when supported by the chosen forensic workflow, can be used to package or manage forensic evidence together with associated case information and integrity metadata. Investigators should verify the format's exact capabilities and compatibility before relying on it. The evidence package should be accompanied by the case identifier, evidence number, source description, device make/model/serial number, acquisition date and time, examiner name, acquisition tool and version, acquisition settings, hash algorithm and values, error logs, and chain-of-custody documentation. Any encryption, compression, or access restrictions should be recorded. The container should be stored in a controlled location with access limited to authorized personnel, and its integrity should be verified before use.

20. Explain data carving and when it may be necessary.

Data carving is the recovery of files or file fragments by identifying recognizable file signatures or structural patterns in raw data, rather than relying entirely on file-system directory entries. It can be useful when directory records have been deleted, file-system metadata is damaged, or data remains in unallocated space. Carving may recover content even when the original file name and folder path are unavailable. Its limitations include fragmented files, false positives, incomplete recovery, and loss of metadata. Recovered items should be validated by checking file structure, content, signatures, and consistency with other evidence. Investigators must not assume that a carved file's apparent name or content proves who created it.

21. Discuss date/time problems and how to document uncertainties.

Digital evidence can contain timestamps in UTC, local time, or a format interpreted according to the system's configured time zone. Devices may have incorrect clocks, daylight-saving changes, unsynchronized time sources, or timestamps that are updated during copying and extraction. Some file systems store several different time attributes, and different applications may display them differently. Investigators should preserve original timestamp values, record the system time zone

and clock state, identify the time source used for normalization, and document any estimated clock offset. A report should distinguish observed times from converted or inferred times. If the available evidence does not permit reliable conversion, the uncertainty should be stated rather than assigning a falsely precise time.

22. Prepare a chain-of-custody record for the laptop from collection through laboratory examination.

A chain-of-custody record should identify the case number, evidence number, item description, serial number, collection location, date and time, collector, packaging and seal number, and reason for collection. Each transfer should record the person releasing the item, the person receiving it, date and time, purpose, and signatures or approved electronic acknowledgements. At laboratory receipt, the examiner should inspect and document the seal condition, compare identifiers, and record any discrepancy. When the item is opened for imaging, the date, operator, reason, and resealing details should be added. Any movement to secure storage, removal for examination, creation of working copies, and return to storage should be logged. The record should be completed contemporaneously; missing information should be explained rather than reconstructed without evidence.

Assignment 2(4)

Case Study: Recovery of Deleted Evidence from Volume Shadow Copies

Introduction and Assumptions

An employee is suspected of deleting documents and browser-related artifacts after an unauthorized activity. A forensic image of the Windows system is available, and Volume Shadow Copies may contain earlier versions of relevant data. The investigation will select relevant evidence, examine it on a forensic copy, validate recovered artifacts, and reconstruct a cautious timeline. The example events below are hypothetical and are not findings from an actual device.

23. Explain how investigators should select and categorize digital evidence before detailed analysis.

Investigators should define the allegation, date range, affected users, file types, and systems before examining the image. They should preserve the complete acquired image and identify evidence sources likely to answer the investigative questions, including user documents, browser artifacts, Windows event logs, application records, deleted-file entries, and Volume Shadow Copies. Evidence can be categorized by relevance, source, reliability, and whether it is direct, corroborating, or contextual. Irrelevant personal data should not be explored beyond the authorized scope. Selection decisions and exclusions should be documented so the examination remains focused and reproducible.

24. Prepare a classification scheme for different categories of files and artifacts.

User-created files include documents, spreadsheets, images, downloads, and project files that may be relevant to the suspected activity. System and application files include operating-system components, browser databases, configuration files, logs, and application caches; they may explain how the system operated or record activity. Potentially suspicious files include unknown executables, scripts, renamed files, or documents linked to the incident; their classification should be based on behavior and context rather than appearance alone. Deleted or recovered artifacts include files recovered from unallocated space, recycle-bin records, or earlier shadow copies. Supporting and contextual evidence includes account information, access permissions, system settings, time-zone details, and routine application records. Each artifact should be labelled with its source, acquisition method, relevance, integrity information, and limitations.

25. Explain how Event Analysis assists in reconstructing activities.

Event Analysis involves organizing event records from Windows, applications, browsers, and other sources into a sequence that can explain system activity. Investigators may review logon events, process or application activity where available, file access and modification records, removable-device events, browser downloads, and deletion-related records. Each event should be interpreted according to what the source actually records. For example, a USB connection event shows that a device was connected, but does not by itself prove that a particular file was copied. Correlating several event types can reveal a more complete sequence. Investigators should also identify gaps, duplicate events, clock differences, and routine system activity that could otherwise be misinterpreted.

26. Explain the purpose of Cloud Analysis and Lead Analysis.

Cloud Analysis examines relevant information stored or processed by cloud services, subject to authorization and provider access. It may help answer whether a file was synchronized, shared, downloaded, or modified through a cloud account, and whether access records or version history are available. Investigators should document the provider, account, collection method, time zone, and limits of the available records. Lead Analysis uses preliminary findings to develop and test investigative leads. For example, a recovered document may identify a file name, user account, cloud folder, or approximate time that guides the next search. A lead is a direction for further investigation, not proof by itself. Both activities should be documented and corroborated with independent evidence.

27. Describe how Volume Shadow Copy analysis can recover or compare earlier evidence.

Volume Shadow Copy Service can preserve point-in-time versions of selected volumes and files. If relevant shadow copies exist in the acquired Windows image, investigators can identify their creation times and examine earlier versions using a compatible forensic tool. They can compare an earlier file with the current or deleted version, calculate hashes, inspect content and metadata, and determine whether the file existed before a suspected deletion. Shadow copies may contain older browser databases or documents, depending on system configuration and what was preserved. They do not necessarily contain every file or every historical version. Investigators should work from the forensic image, document the extraction method, and avoid implying that absence from a shadow copy proves a file never existed.

28. Construct a hypothetical timeline of file creation, modification, copying, deletion, and recovery.

For illustration, assume a document named “exam_notes.docx” appears in a user’s Documents folder on 12 September at 10:05 local time. At 10:18, metadata indicates that the document was modified. At 10:24, a USB device is connected to the computer. At 10:27, a relevant file record or other artifact suggests that a copy may have been created on the USB. At 10:41, the original document is deleted. On 13 September, investigators identify a Volume Shadow Copy created before the deletion and recover an earlier version of the document from it. These times are hypothetical and would require verification from actual artifacts. The USB connection alone would not prove copying, and the recovered earlier version would establish prior existence but not necessarily identify the person who created or deleted the file.

29. Explain how the recovered artifact should be validated.

The examiner should record the exact source, shadow-copy identifier, extraction method, tool version, and recovery time. Calculate a hash for the recovered artifact and preserve the extracted file without further modification. Check whether the file structure and content are consistent with its stated type, and compare it with other versions using hashes, metadata, or content differences. Verify that the source shadow copy existed in the acquired image and document its timestamp and any uncertainty. Correlate the artifact with user profiles, event logs, browser or USB records, and the wider timeline. If a file is only partially recovered, has conflicting timestamps, or cannot be

opened reliably, the report should disclose that limitation. The examiner should retain sufficient notes and logs for independent review.

30. Identify possible errors or alternative explanations affecting timestamps and deleted artifacts.

Timestamps may be affected by time-zone conversion, clock drift, daylight-saving settings, copying between devices, or deliberate manipulation. A shadow copy's creation time is not necessarily the exact time a particular file was created. A recovered file may be an older version, a duplicate, or an incomplete remnant rather than the precise file that was deleted. Automated backup, synchronization, antivirus scanning, or an application's normal behavior may explain some changes. Deleted data may be overwritten or altered by normal system use, and forensic tools can interpret damaged structures differently. Investigators should validate findings, preserve original values, compare multiple sources, and state uncertainty clearly.

Assignment 2(5)

Case Study: Multi-Source Investigation of a Suspected Cyber Incident

Introduction and Assumptions

A student account is suspected of being used to access an online service without authorization. Potential evidence includes a Windows computer, browser artifacts, email and messaging records, a mobile phone, and Internet or cloud records. The investigation must be authorized, preserve evidence from each source, correlate the records carefully, and distinguish account activity from proof of an individual's intent or identity.

31. Identify potential evidence from Windows and other operating systems.

On Windows, relevant sources may include Security and System event logs, user profiles, file-system metadata, recent-file records, application execution artifacts, PowerShell or command history where retained, USB connection records, installed applications, and network configuration. On macOS, useful sources may include unified logs, file metadata, browser data, user account records, and application artifacts. On Linux, investigators may examine authentication logs, journal logs, shell history, file metadata, scheduled tasks, and network configuration. Other operating systems may retain application, account, and device logs. The exact evidence available depends on system settings, retention, permissions, encryption, and whether the records were overwritten or deleted. All sources should be acquired through authorized and documented methods.

32. Explain what browser artifacts may reveal and why they require careful interpretation.

Browser artifacts can include history, downloads, cache entries, cookies, saved sessions, autofill data, bookmarks, site permissions, and timestamps. Depending on the browser and configuration, they may help establish that a browser profile interacted with a website or downloaded a file. However, history can be deleted, private-browsing sessions may leave limited records, and synchronization can place activity on multiple devices. A cached item does not always show that the user deliberately viewed it, and a URL in history does not prove that a transaction was completed. Multiple people may use the same profile, and malware or automated processes can generate traffic. Investigators should preserve the browser database, document the profile and time settings, and corroborate key events with server, email, or device records.

33. List evidence available from email and messaging systems and explain correlation.

Email evidence may include message bodies, headers, sender and recipient addresses, message identifiers, attachments, timestamps, delivery information, and account login alerts. Messaging systems may retain conversation text, participant identifiers, media files, delivery/read status, call records, and local databases, depending on the service and collection method. Message headers and server-side records can help distinguish a message's displayed sender from the systems through which it was transmitted. Investigators can correlate an attachment's hash with a downloaded file, compare message timestamps with browser history, and compare account alerts with Windows or mobile login activity. Message content should be interpreted in context, and screenshots alone may

not preserve all metadata or prove authenticity. Collection must respect authorization and applicable privacy requirements.

34. Explain forensic considerations and challenges when examining a mobile phone.

Before examining a phone, record its make, model, serial or device identifier where appropriate, operating system, visible condition, power state, lock state, SIM and memory-card details, and network connections. The examiner should consider whether the phone is encrypted and whether powering it off or allowing its battery to drain could make data inaccessible. Isolation from networks may be appropriate to prevent remote wiping or synchronization, but the method and effects should be documented. Use an authorized acquisition method suitable for the device and preserve tool logs. Mobile evidence may include call logs, messages, app databases, photos, location records, browser data, and notification history. Challenges include encryption, locked bootloaders, vendor restrictions, app-level encryption, cloud-only content, rapidly changing data, and differences between logical and physical acquisition. A recovered artifact should be linked to the device and interpreted with its limitations.

35. Discuss Internet and cloud-based evidence relevant to the investigation.

Potential sources include service login history, IP addresses, authentication and multi-factor authentication records, account recovery events, cloud file access logs, synchronization history, version history, shared-link activity, DNS records, firewall logs, and records held by an Internet service provider. These sources may help identify the time, account, device, or network associated with an event. IP addresses generally identify a network endpoint or public connection, not necessarily a particular person or device. Shared networks, VPNs, proxies, carrier-grade NAT, and dynamic address assignment can complicate attribution. Cloud records may be retained only for a limited period and may require formal legal or provider procedures. Investigators should preserve records promptly, document how they were obtained, normalize timestamps, and correlate them with local device evidence.

36. Explain cross-source correlation using at least five hypothetical artifacts.

Assume a hypothetical investigation finds the following related artifacts. First, a Windows log records an account logon at 19:02. Second, browser history shows a visit to the affected service at 19:07. Third, an email alert from the service reports a new login at 19:08. Fourth, a mobile phone contains a notification about a one-time verification request at 19:09. Fifth, a cloud audit record shows a file request at 19:11 from an IP address associated with the same session. Sixth, a message sent at 19:15 discusses the file involved. Investigators would compare account identifiers, device details, IP addresses, timestamps, message identifiers, and any file hashes. This hypothetical pattern may support a connected sequence, but each artifact needs authentication and validation. A notification or message alone does not prove who initiated the login, and the shared IP address may represent multiple users.

37. Explain at least six major challenges in multi-source digital forensics.

One challenge is different timestamp formats and clock accuracy across devices and providers. A second is encryption, which may prevent access to mobile, computer, or cloud content. A third is data volatility and short retention periods, which can cause evidence to disappear. A fourth is the

large volume of data across devices and services, making relevant artifacts difficult to identify. A fifth is inconsistent formats and proprietary applications that require different tools. A sixth is cloud jurisdiction and legal access requirements, which may restrict collection. A seventh is synchronization and duplication, which can make one event appear on several devices. An eighth is attribution: a record may link an event to an account or IP address without identifying the person responsible. A ninth is deleted, incomplete, or corrupted data. A tenth is the risk of altering evidence during collection. These challenges require a scoped plan, validated tools, detailed notes, independent corroboration, and transparent reporting of limitations.

38. Prepare a final conclusion distinguishing direct evidence, corroborating evidence, and unresolved questions.

Based on the hypothetical scenario, direct technical evidence would include authenticated service records that document a specific account event, if the records are validated and reliably tied to the incident. Corroborating evidence could include browser history, a matching service email alert, mobile notification records, cloud audit logs, and relevant messages whose timestamps and identifiers agree. Together, these sources may support the conclusion that the account was used to access the service at the stated time. However, the evidence may not establish who physically controlled the device, whether credentials were compromised, whether a shared network was used, or whether all timestamps are accurate. These remain unresolved questions unless further evidence answers them. The final report should avoid declaring the student responsible solely because their account or device appears in the records. It should state what is established, what is supported but not conclusive, and what additional evidence would be required.

Overall Conclusion

Digital forensics depends on preserving evidence, documenting every action, examining verified copies, and interpreting artifacts in context. Across these five case studies, reliable conclusions require proper authorization, chain of custody, integrity verification, careful timestamp handling, and correlation between independent sources. A file's presence, a login record, or an account identifier may be important, but none should automatically be treated as proof of a particular person's intent. Investigators should report both supporting evidence and limitations so that the findings remain transparent, repeatable, and defensible.